

# **Logarithm and Exponentiation**

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# 1 Logarithm (對數) and Exponentiation (指數)

## I Definitions

### i Principal branch (主分支)

In the following definitions,  $\arg(x \in \mathbb{C}_{\neq 0})$  represents the argument (輻角) of  $x$ . If we replace  $\arg(x)$  with the principal argument (輻角主值) of  $x$ , we get the definition of the principal branch of them.

### ii Positive integer exponent (正整數指數)

The exponent of  $a \in \mathbb{R}$  to the power of  $n \in \mathbb{N}$ , denoted as  $a^n$ , is defined as

$$a^n := \prod_{k=1}^n a.$$

### iii Natural logarithm (自然對數) in real numbers

The natural logarithm of  $x$ , denoted as  $\ln(x)$  or  $\log_e(x)$ , is defined in real numbers as:

$$\ln(x) := \int_1^x \frac{1}{t} dt, \quad x \in \mathbb{R}_{>0}.$$

### iv Natural logarithm in complex numbers

The natural logarithm of  $x$ , denoted as  $\ln(x)$  or  $\log_e(x)$ , is defined in complex numbers as:

$$\ln|x| + i \arg(x), \quad x \in \mathbb{C}_{\neq 0}.$$

### v Logarithm (對數) in real numbers

The logarithm of  $x$ , called argument (真數), to the  $a$ , called base (底數), denoted as  $\log_a(x)$ , is defined in real numbers as

$$\log_a(x) := \frac{\ln(x)}{\ln(a)}, \quad a, x \in \mathbb{R}_{\neq 0} \wedge a \neq 1.$$

### vi Logarithm in complex numbers

The logarithm of  $x$ , called argument, to the  $a$ , called base, denoted as  $\log_a(x)$ , is defined in complex numbers as

$$\log_a(x) := \frac{\ln(x)}{\ln(a)}, \quad a, x \in \mathbb{C}_{\neq 0} \wedge a \neq 1.$$

### vii Common logarithm (常用對數)

$\log_{10}(x)$  is called the common logarithm of  $x$ , also denoted as  $\log(x)$ .

### viii Exponent (指數) in real numbers

The exponent of  $a$ , called the base (底數), to the exponent (指數) or power (冪次) of  $n$ , denoted as  $a^n$ , is defined in real numbers as

$$a^n := \begin{cases} e^{n \ln(a)}, & a \in \mathbb{R}_{>0} \wedge n \in \mathbb{R}, \\ -|a|^n, & a \in \mathbb{R}_{<0} \wedge \exists p, q \in \mathbb{N} \text{ s.t. } n = \frac{2p-1}{2q-1}, \\ |a|^n, & a \in \mathbb{R}_{<0} \wedge \exists p, q \in \mathbb{N} \text{ s.t. } n = \frac{2p-2}{2q-1}, \\ 0, & a = 0 \wedge n \in \mathbb{R}_{>0}. \end{cases}$$

Exponentiation is right-associative, that is,

$$a^{b^c} = a^{(b^c)}, \quad \text{not } (a^b)^c.$$

### ix Euler's number (歐拉數/尤拉數)

The Euler's number, denoted as  $e$ , is defined as

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

Some other equivalent definitions include:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

*Proof.*

$$\begin{aligned} \left(1 + \frac{1}{n}\right)^n &= \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k} \\ &= \sum_{k=0}^n \frac{\prod_{j=0}^{k-1} (n-j)}{n^k k!} \\ &= \sum_{k=0}^n \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right) \frac{1}{k!} \end{aligned}$$

For fixed  $k$ :

$$\begin{aligned} \lim_{n \rightarrow \infty} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right) &= 1 \\ \lim_{n \rightarrow \infty} \sum_{k=0}^n \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right) \frac{1}{k!} &= \sum_{k=0}^{\infty} \frac{1}{k!}. \end{aligned}$$

□

$$e^x := \text{Solution of } f'(x) = f(x), \quad f(0) = 1,$$

that is,

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$$

*Proof.* By separation of variables:

$$\begin{aligned}\frac{dy}{dx} &= y \\ \frac{dy}{y} &= dx. \\ \ln|y| &= x + C \\ y(x) &= Ce^x \\ C &= 1 \\ y(x) &= e^x.\end{aligned}$$

□

$$\int_1^e \frac{dx}{x} = 1.$$

*Proof.*

$$\begin{aligned}f(x) &:= \sum_{n=0}^{\infty} \frac{x^n}{n!}. \\ f'(x) &= \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = f(x).\end{aligned}$$

By uniqueness of solution of  $f'(x) = f(x)$  and  $f(0) = 1$ ,

$$f(x) = e^x.$$

Therefore,

$$e = f(1).$$

By the definition of natural logarithm:

$$\begin{aligned}\ln y &= \int_1^y \frac{dx}{x}. \\ \int_1^e \frac{dx}{x} &= 1.\end{aligned}$$

□

## x Euler's formula (歐拉公式)

For any real number  $\theta$ :

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

## xi Exponent in complex numbers

The exponent of  $a$ , called the base, to the exponent or power of  $n$ , denoted as  $a^n$ , is defined in complex numbers as

$$a^n := \begin{cases} e^{n \ln(a)}, & a \in \mathbb{C}_{\neq 0} \wedge n \in \mathbb{C}, \\ 0, & a = 0 \wedge n \in \mathbb{R}_{>0}. \end{cases}$$

Exponentiation is right-associative, that is,

$$a^{b^c} = a^{(b^c)}, \quad \text{not } (a^b)^c.$$

## **xii Root (根號)**

The  $y$ th root ( $y$  次方根) of  $w \in \mathbb{C}$  and  $y$  such that  $w^{\frac{1}{y}}$  is defined, denoted as  $\sqrt[y]{w}$  is defined as the principal branch of  $w^{\frac{1}{y}}$ , where  $\sqrt{\phantom{x}}$  is called the radical symbol, radical sign, root symbol, or surd (根號).

## **xiii Scientific notation (科學記號)**

Scientific notation refers to the representation of a real number in the form of  $a \times 10^n$ , where  $1 \leq |a| < 10 \wedge n \in \mathbb{Z}$ .

## **xiv Characteristic (首數) and mantissa (尾數)**

Given a common logarithm  $\log(x)$ , the characteristic of it is defined as  $\lfloor \log(x) \rfloor$ , and the mantissa of it is defined as  $\log(x) - \lfloor \log(x) \rfloor$ .

## **xv Logarithmic function (對數函數)**

$f(x) = k \log_a(x)$  where  $a > 0 \wedge a \neq 1 \wedge k \in \mathbb{R}_{\neq 0}$  is called a logarithmic function with base  $a$ , of which the domain is  $\mathbb{R}_{>0}$  and the range is  $\mathbb{R}$ .

## **xvi Exponential function (指數函數)**

$f(x) = ka^x$  where  $a \neq 0 \wedge k \in \mathbb{R}_{\neq 0}$  is called an exponential function with base  $a$ , of which the domain is  $\mathbb{R}$  and the range is  $\mathbb{R}_{>0}$  when  $k > 0$  and  $\mathbb{R}_{<0}$  when  $k < 0$ .

## **xvii Natural exponential function (自然指數函數)**

$e^x$  is called the natural exponential function.

## **xviii Exponential growth (指數成長)**

Exponential growth refers to a real function such that  $f'(x) = kf(x)$  where  $k \in \mathbb{R}_{\neq 0}$  (or  $k \in \mathbb{R}_{>0}$  in some contexts), that is,  $f(x) = e^{kx} + c$  where  $c \in \mathbb{R}$ . And the equation  $f'(x) = kf(x)$  is sometimes called the law of natural growth.

Exponential decay refers to a real function such that  $f'(x) = kf(x)$  where  $k \in \mathbb{R}_{<0}$ , that is,  $f(x) = e^{kx} + c$  where  $c \in \mathbb{R}$ . And the equation  $f'(x) = kf(x)$  is sometimes called the law of natural decay.

## II Laws

### i Logarithmic laws (對數律)

$$\log_a(r) + \log_a(s) = \log_a(rs)$$

$$\log_a(r) - \log_a(s) = \log_a\left(\frac{r}{s}\right)$$

$$\log_{a^m}(r^n) = \frac{n}{m} \log_a(r)$$

$$\log_a(b) = \frac{\log_c(b)}{\log_c(a)}$$

$$a^{\log_c(b)} = b^{\log_c(a)}$$

### ii Exponential laws (指數律)

$$a^r \cdot a^s = a^{r+s},$$

$$(a^r)^s = a^{rs},$$

$$(a \cdot b)^r = a^r \cdot b^r$$

### iii Inverse

For any  $a > 0 \wedge a \neq 1 \wedge b \neq 0$ , the logarithmic function  $k \log_a\left(\frac{x}{b}\right)$  and the exponential function  $ba^{\frac{x}{k}}$  are inverses of each other, that is, they are symmetric about  $y = x$ .

### iv Tangent

In the  $xy$  plane,  $a = e^{(e^{-1})}$  if and only if any two of  $y = a^x$ ,  $y = x$ , and  $y = \log_a x$  are tangent to each other. If  $a = e^{(e^{-1})}$ ,  $y = a^x$ ,  $y = x$ , and  $y = \log_a x$  are tangent at  $(e, e)$ , and any two of them do not meet at any point other than  $(e, e)$ .

*Proof.*

$$\frac{da^x}{dx} = \ln(a)a^x = 1$$

$$x = a^x = \frac{1}{\ln(a)}$$

$$a^{\frac{1}{\ln(a)}} = \frac{1}{\ln(a)}$$

$$\ln(a) \frac{1}{\ln(a)} = 1 = -\ln(\ln(a))$$

$$a = e^{(e^{-1})}$$

□

### v Self-power function

$$\arg \min(x^x) = \frac{1}{e}, \quad \min(x^x) = (e^{-1})^{(e^{-1})}$$

### III Values

$$\log 2 \approx 0.3010, \quad \log e \approx 0.4343, \quad \log 3 \approx 0.4771, \quad \log \pi \approx 0.4971, \quad \log 4 \approx 0.6021, \quad \log 5 \approx 0.6990, \\ \log 6 \approx 0.7782, \quad \log 7 \approx 0.8451, \quad \log 8 \approx 0.9031, \quad \log 9 \approx 0.9542, \quad \log 11 \approx 1.0414, \quad \log 12 \approx 1.0792.$$

$$\ln 2 \approx 0.6931, \quad \ln 3 \approx 1.0986, \quad \ln \pi \approx 1.1447, \quad \ln 4 \approx 1.3863, \quad \ln 5 \approx 1.6094, \quad \ln 6 \approx 1.7918,$$

$$\ln 7 \approx 1.9459, \quad \ln 10 \approx 2.3026, \quad \ln 11 \approx 2.3980, \quad \ln 12 \approx 2.4849, \quad \frac{1}{\ln 2} \approx 1.4427.$$

$$\sqrt{2} \approx 1.4142, \quad \sqrt{e} \approx 1.6487, \quad \sqrt{3} \approx 1.7321, \quad \sqrt{\pi} \approx 1.7725, \quad \sqrt{5} \approx 2.2361, \\ \sqrt{6} \approx 2.4495, \quad \sqrt{7} \approx 2.6458, \quad \sqrt{10} \approx 3.1623, \quad \sqrt{11} \approx 3.3166, \quad \sqrt{13} \approx 3.6056, \\ \sqrt{14} \approx 3.7417, \quad \sqrt{15} \approx 3.8730, \quad \sqrt{17} \approx 4.1231, \quad \sqrt{19} \approx 4.3590, \quad \sqrt{21} \approx 4.5826, \\ \frac{\sqrt{5}+1}{2} \approx 1.6180, \quad \frac{\sqrt{5}-1}{2} \approx 0.6180, \quad \frac{\sqrt{6}+\sqrt{2}}{4} \approx 0.9659, \quad \frac{\sqrt{6}-\sqrt{2}}{4} \approx 0.2588, \\ \frac{\sqrt{10+2\sqrt{5}}}{4} \approx 0.9511, \quad \frac{\sqrt{10-2\sqrt{5}}}{4} \approx 0.5878.$$

$$\langle a_n = n^2 \rangle_{n=1}^{20} = 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400.$$

$$\langle a_n = n^2 \rangle_{n=21}^{30} = 441, 484, 529, 576, 625, 676, 729, 784, 841, 900.$$

$$\langle a_n = n^2 \rangle_{n=31}^{40} = 961, 1024, 1089, 1156, 1225, 1296, 1369, 1444, 1521, 1600.$$

$$\langle a_n = n^{-1} \rangle_{n=1}^{15} = 1, 0.5, 0.\bar{3}, 0.25, 0.2, 0.1\bar{6}, 0.\overline{142857}, 0.125, 0.\bar{1}, 0.1, 0.\overline{09}, 0.08\bar{3}, 0.\overline{076923}, 0.07\overline{14285}, 0.0\bar{6}.$$

$$\frac{1}{\sqrt{2}} \approx 0.7071, \quad \frac{1}{\sqrt{3}} \approx 0.5774, \quad \frac{1}{\sqrt{5}} \approx 0.4772, \quad \frac{1}{2\sqrt{2}} \approx 0.3536, \quad \frac{1}{2\sqrt{3}} \approx 0.2887, \quad \frac{1}{2\sqrt{5}} \approx 0.2236.$$

$$e \approx 2.7183, \quad \frac{1}{e} \approx 0.3680, \quad e^e \approx 15.1543, \quad e^{(e^{-1})} \approx 1.4447, \quad (e^{-1})^e \approx 0.0660, \quad (e^{-1})^{(e^{-1})} \approx 0.6922.$$